

AI Benchmark for Molecular Identification of Anopheles Mosquitoes

Conventional PCR + agarose gel electrophoresis | Life Sciences AI evaluation case study

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Domain	Molecular biology / malaria vector surveillance
Benchmark type	Synthetic, scientifically grounded evaluation task
Core skills demonstrated	Methodological rigor, protocol fidelity, data interpretation, source synthesis, rubric design

Project summary. I designed a compact AI benchmark that tests whether a model can interpret a species-diagnostic conventional PCR experiment for *Anopheles gambiae* s.s. and *Anopheles arabiensis*. The model receives sample metadata, primer and amplicon references, run controls, a synthetic agarose gel, and a protocol. It must validate the experiment first, classify interpretable specimens, flag ambiguous results, and avoid unsupported conclusions.

Why this is a useful benchmark

- It cannot be solved by simple band spotting: the model must reason about positive controls, the no-template control, band size, band quality, and uncertainty.
- The dataset contains clean, weak, absent, nonspecific, smeared, and dual-band outcomes.
- The gold answer is separated from the task artifacts so the same case can be used to evaluate different AI responses.
- The rubric rewards scientific reasoning and penalizes confident conclusions that exceed the evidence.

Portfolio disclosure: all specimen metadata and gel results are synthetic. The assay logic, primer sequences, and expected product sizes are adapted from published PCR literature.

1. Scientific basis and benchmark protocol

Members of the *Anopheles gambiae* complex are closely related sibling species. The classic Scott et al. PCR uses species-specific sequences in the ribosomal DNA intergenic spacer to distinguish members of the complex. In this benchmark, a single clear product at approximately 390 bp supports *An. gambiae* s.s., while a single clear product at approximately 315 bp supports *An. arabiensis*, provided the run controls are valid.

Item	Benchmark reference
Universal primer	5'-GTGTGCCCCTTCCTCGATGT-3'
GA primer	5'-CTGGTTTGGTCGGCACGTTT-3'
AR primer	5'-AAGTGTCTTCTCCATCCTA-3'
Expected product	<i>An. gambiae</i> ~390 bp; <i>An. arabiensis</i> ~315 bp
Cycling profile	95 C 1 min; 35 cycles of 94 C 25 s, 50 C 25 s, 72 C 30 s; final extension 72 C 5 min
Gel	2% agarose; compare products with molecular-weight ladder
Run QC	G+ and A+ must show expected bands; NTC must remain free of visible target-sized amplification

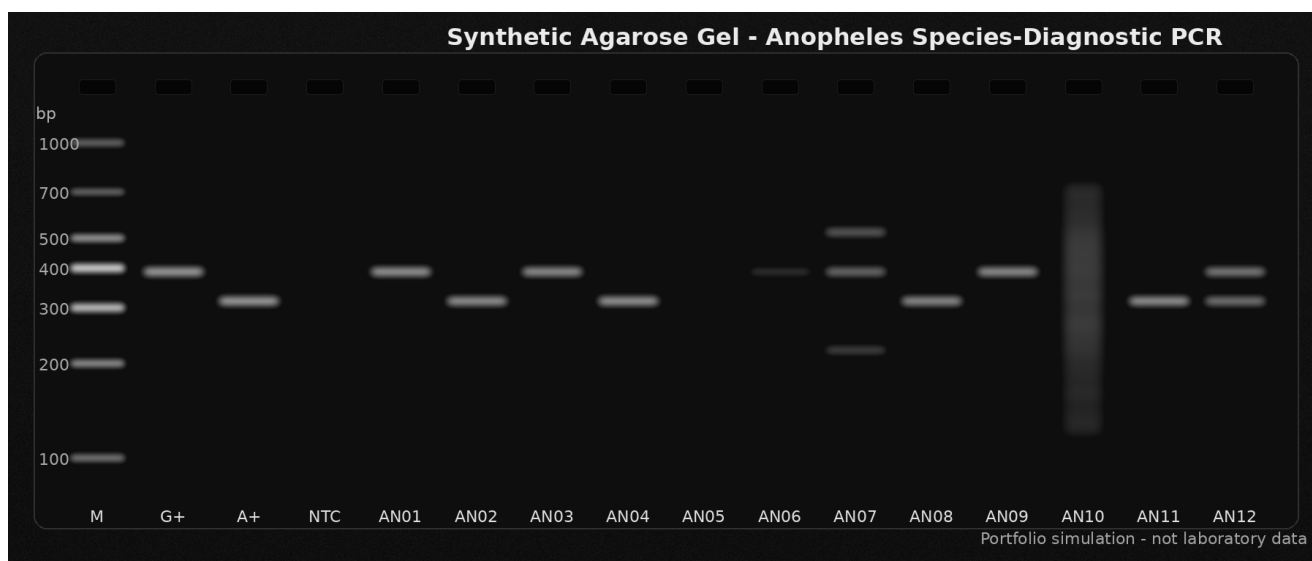
Reasoning rule used in the benchmark

1	Validate controls
2	Locate the specimen band relative to the ladder
3	Assess whether amplification is clean, weak, nonspecific, smeared, or absent
4	Assign a species only when the evidence supports it
5	State uncertainty and recommend follow-up for problematic lanes

This protocol is a simulation for benchmark construction, not a laboratory SOP or diagnostic instruction.

2. Synthetic experimental dataset

The overall run is deliberately valid: the *An. gambiae* and *An. arabiensis* positive controls amplify at their expected sizes and the no-template control is clean. The challenge is therefore to decide which individual specimen results are trustworthy.



Sample	Observed result	Gold classification	Why it matters
AN01	Strong ~390 bp	<i>An. gambiae</i>	Clean species call
AN02	Strong ~315 bp	<i>An. arabiensis</i>	Clean species call
AN05	No visible band	Inconclusive	No band is not proof of species absence
AN06	Weak ~390 bp	Probable <i>An. gambiae</i>	Correct size but reduced confidence
AN07	~390 + extra products	Inconclusive	Nonspecific amplification
AN10	Diffuse smear	Inconclusive	Reliable size cannot be assigned
AN12	~390 + ~315 bp	Dual-band / unresolved	Do not force a single-species call

Key benchmark principle: a valid PCR run does not make every specimen result valid. Individual lanes can still be inconclusive or require repeat testing.

3. Gold-standard reasoning

The gold answer is not simply a list of species labels. Each call must be justified from the controls, expected product size, and quality of amplification.

AN05 - no amplification

The correct response is inconclusive, not "neither species." The run controls pass, but this specimen has no internal amplification control. Low DNA quantity, poor template quality, PCR inhibition, or reaction failure remain possible. Repeat the PCR and investigate DNA quality.

AN06 - weak 390 bp band

The size is consistent with *An. gambiae*, so the result is probable *An. gambiae*. However, weak amplification lowers confidence. A careful evaluator recommends confirmation rather than reporting it with the same certainty as a strong clean band.

AN07 - multiple products

Multiple discrete bands indicate nonspecific amplification. Even though one product lies near the *gambiae*-sized position, a clean single-species call is not justified. Repeat after troubleshooting reaction specificity.

AN10 - smear

A smear prevents reliable amplicon sizing and is therefore inconclusive. Possible causes include poor or degraded DNA, excess template, contaminants, or suboptimal PCR conditions. The correct action is repeat/troubleshoot, not species assignment.

AN12 - 390 and 315 bp products

Both expected species-sized products are visible. The evidence supports a dual-band phenotype, but not a definitive explanation. Mixed template, a hybrid signal, or contamination are possibilities that require confirmation. The model should not invent one cause as certain.

What this tests in an AI system

The benchmark rewards models that distinguish observation from interpretation, state uncertainty explicitly, and recommend a scientifically appropriate next step. It penalizes models that equate "any band" with a positive result or that force every specimen into a binary species label.

4. Evaluation rubric and portfolio value

Evaluation dimension	Points	What full credit requires
Run-control validation	4	Checks G+, A+, and NTC before interpreting specimens; correctly concludes that the overall run is valid.
Clean species calls	4	Correctly identifies clear ~390 bp and ~315 bp specimens.
No/weak amplification	4	Handles AN05 and AN06 with appropriate uncertainty and follow-up.
Nonspecific/smear/dual band	6	Correctly handles AN07, AN10, and AN12 without overclaiming.
Source and protocol fidelity	1	Uses only supported assay logic and does not invent thresholds.
Scientific communication	1	Clear, concise, evidence-linked explanation.
TOTAL	20	Passing target for a high-quality response: 16/20 or above.

Skills demonstrated

Scientific	AI evaluation / research operations
Conventional PCR and gel interpretation	Benchmark task design
Experimental controls and quality assurance	Gold-standard answer creation
Protocol interpretation	Atomic scoring-rubric design
Data-quality troubleshooting	Source fidelity and uncertainty handling

Selected sources

Scott JA, Brogdon WG, Collins FH (1993). Identification of single specimens of the *Anopheles gambiae* complex by the polymerase chain reaction. PubMed: <https://pubmed.ncbi.nlm.nih.gov/8214283/>

Walker et al. Field genotyping study showing predicted ~390 bp *An. gambiae* and ~315 bp *An. arabiensis* products and examples of nonreactive/dual-band specimens. <https://pmc.ncbi.nlm.nih.gov/articles/PMC1808465/>

Published western Kenya PCR workflow. Primer sequences and conventional PCR/gel implementation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2980855/>